

ChE 201 – Introduction to Thermodynamics –Spring 2018

Student Learning Outcome 1

Rubric for assessment problem on final exam (Problem #4)

Application of energy and entropy balances to a piston/cylinder device with water.

SLO 1 Performance Indicator (PI)	(1) Unsatisfactory	(2) Developing	(3) Satisfactory
1. Apply mathematical and scientific principles to formulate models and systems relevant to the subject Problem 4a, 4b 6 pts	Does not understand the connection between mathematical models and the system or process to be analyzed or designed (<3 pts)	Chooses a mathematical model or scientific principle that applies to an engineering problem, but has trouble in model development (3 pts)	Able to successfully combine mathematical and/or scientific principles to formulate models and systems relevant to chemical engineering (4-6 pts)
2. Solve engineering problems by using the concepts of algebra, calculus, differential equations and/or linear algebra Problem 4c, 4pts	Does not understand the application of algebra, calculus, differential equations and/or linear algebra in solving chemical engineering problems (<2 pts)	Shows nearly complete understanding of applications of calculus and/or linear algebra in problem-solving (2 pts)	Applies concepts of algebra, calculus, differential equations and/or linear algebra to solve chemical engineering problems (3-4 pts)
4. Translates academic theory into engineering applications Problem 4d, 4e, 4f 20 pts	Does not appear to grasp the connection between theory and the problem (< 8 pts)	Some gaps in understanding the application of theory to the problem and expects theory to predict reality (9-12 pts)	Translates academic theory into engineering applications and accepts limitations of mathematical models of physical reality (13-20 pts)

Averages:

Performance Indicators: (1) 2.71 (2) 2.57 (4) 2.33 Overall Average: 2.54

Average scores were fairly uniform although there was wide variation among students. Clearly, most students were strong in formulating the engineering system and model equations, and a majority 13/21 were satisfactory or higher in applying the engineering principles to solve the three problems.

QUESTION 4: (30 pts)

A piston-cylinder device which contains 3 kg of saturated water vapor at 6 bar. Heat is transferred between the piston-cylinder and a 25°C thermal reservoir until the saturated water vapor becomes a saturated liquid at 6 bar. The piston moves freely and maintains constant pressure throughout the process. Kinetic and potential energy can be ignored. Steam Tables are at the end of Exam.

- a) (2 pts) Draw a schematic of the system and label the system boundary.
- b) (4 pts) State the engineering model
- c) (4 pts) Write the simplified energy balance and simplified entropy balance.
- d) (10 pts) Determine the work in **kJ**, and the heat transferred in **kJ**.
- e) (5 pts) Determine the change in entropy of the water during the process, **ΔS , in kJ/K**
- f) (5 pts) Determine the entropy generated during the process, **σ , in kJ/K**.

Solution

Solution SET

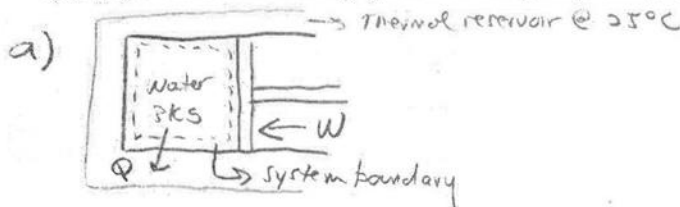
CHE 201 Final Exam

May 10th, 2018

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- (8 pts) Draw a schematic of the system and label the system boundary.
- (4 pts) State the engineering model
- (8 pts) Write the simplified energy balance and simplified entropy balance.
- (10 pts) Determine the work in kJ, and the heat transferred in kJ.
- (5 pts) Determine the change in entropy of the water during the process, ΔS , in kJ/K
- (5 pts) Determine the entropy generated during the process, σ , in kJ/K.



- 1) Water in a piston-cylinder assembly is the closed system
- 2) $\Delta KE + \Delta PE$ are ignored
- 3) Heat transfer to thermal reservoir
- 4) Volume change is the work mode

c) Energy balance: $\Delta U + \Delta KE + \Delta PE = Q - W$
 $\Delta U = Q - W = m(u_2 - u_1) = Q - W$

Entropy Balance: $\Delta S = \int_1^2 \left(\frac{\delta Q}{T} \right) + \sigma_{cv}$

d) $W_{\text{work}} (\text{kJ}) = \int_1^2 P dV$ SAT vapor $\rightarrow$ SAT Liquid
 Constant $P = 6 \text{ bar}$

$W_{\text{work}} (\text{kJ}) = mP(v_2 - v_1)$ Table A-3

$W_{\text{work}} (\text{kJ}) = (3 \text{ kg})(6 \text{ bar})(1.1008 \times 10^{-3} - 0.3157) \frac{\text{m}^3}{\text{kg}}$ $v_2 = 1.1008 \times 10^{-3} \frac{\text{m}^3}{\text{kg}}$
 $v_1 = 0.3157 \frac{\text{m}^3}{\text{kg}}$

$= (18)(-0.3146) \text{ bar} \cdot \text{m}^3$ $m = 3 \text{ kg}$

$= -5.66 \text{ bar} \cdot \text{m}^3$ $\frac{10^5 \text{ N/m}^2}{1 \text{ bar}} \frac{1 \text{ kJ}}{10^3 \text{ N} \cdot \text{m}} = \boxed{-566.3 \text{ kJ}}$
 ON system
 $= W_{\text{work}}$

Heat (kJ): $Q = m(u_2 - u_1) + W$

$u_2 = 669.90 \frac{\text{kJ}}{\text{kg}}$ $Q (\text{kJ}) = (3 \text{ kg})(669.9 - 2567.4) \frac{\text{kJ}}{\text{kg}} + (-566.3 \text{ kJ})$
 $u_1 = 2567.4$ $Q = (3(-1897.5) - 566.3) \text{ kJ}$
 $Q = -5692.5 - 566.3 = \boxed{-6258.8 \text{ kJ} = Q}$
 From system

e) $\Delta S = ?$ $s_1 = 6.76 \frac{\text{kJ}}{\text{kg} \cdot \text{K}}$ $\Delta S = (3 \text{ kg})(1.9312 - 6.760) \frac{\text{kJ}}{\text{kg} \cdot \text{K}}$
 $s_2 = 1.9312$ $= (3)(-4.8288) \frac{\text{kJ}}{\text{K}}$
 $\Delta S = \boxed{-14.4864 \frac{\text{kJ}}{\text{K}}}$

f) $\sigma_{\text{cv}} = \Delta S - \left(\frac{Q}{T}\right)_b$ $T_b = 25^\circ\text{C} + 273.15 \text{ K} = 298.15 \text{ K}$

$\sigma_{\text{cv}} = -14.4864 \frac{\text{kJ}}{\text{K}} - \left(\frac{-6258.8 \text{ kJ}}{298.15 \text{ K}}\right) =$

$= -14.4864 \frac{\text{kJ}}{\text{K}} + 20.992 \frac{\text{kJ}}{\text{K}} = \boxed{6.5056 \frac{\text{kJ}}{\text{K}} = \sigma_{\text{cv}}}$

$\sigma_{\text{cv}} > 0$ possible!